Seeing Through the Clutter: Disentangling Distractors in Vision-Language Models

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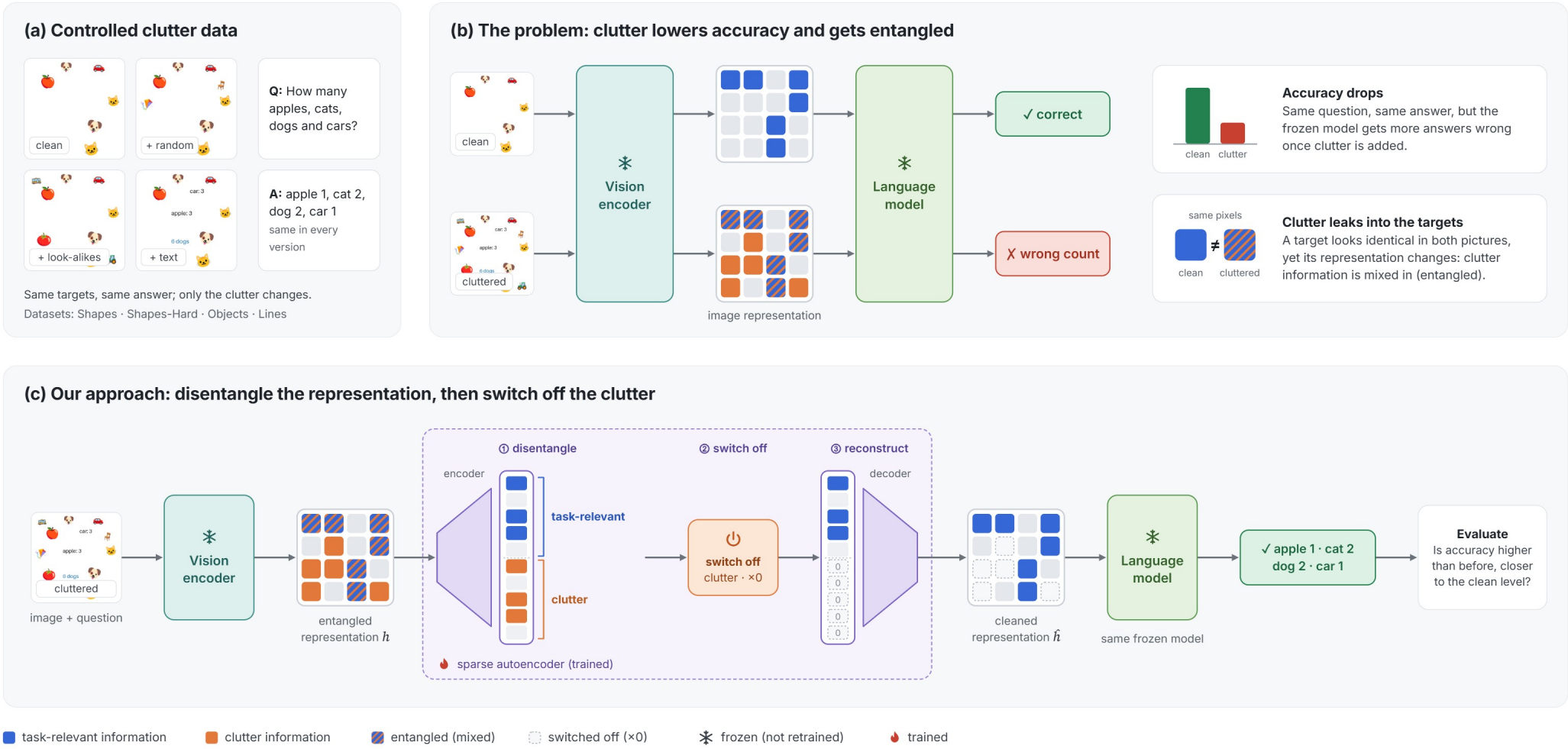


Figure 1. Overview of the project. (a) Each scene is drawn clean and with different clutter (random objects, look-alikes, misleading text); the targets, question and answer stay identical, so any change in behaviour is caused by the clutter. (b) We measure how much clutter lowers a frozen VLM's accuracy and test whether it leaks into the representation of the unchanged targets (entanglement). (c) A sparse autoencoder splits the entangled representation into features; we switch off the clutter features, rebuild the representation and let the same frozen model answer, aiming for higher accuracy than without the fix.

1 Problem and hypothesis

Vision-language models (VLMs) [1] handle clean images well but often fail when the image also contains clutter: extra objects, look-alikes or misleading text. We ask why, and how to fix it without retraining.

Hypothesis. The model's image representation entangles clutter with the information the answer needs, and this causes wrong answers. Separating the two and switching off the clutter part should raise accuracy.

- **RQ1 Effect.** How much does each kind of clutter lower accuracy, and which kinds hurt most?
- **RQ2 Entanglement.** Does clutter leak into the representation of unchanged targets and cause errors?
- **RQ3 Disentanglement.** Can a sparse autoencoder [2, 3] isolate the clutter, and does removing it help?

2 Datasets

Four synthetic datasets of 500 scenes each (12,500 images, split by scene 350 / 75 / 75). Targets are pixel-identical in every version, so the answer never changes and any accuracy drop is caused by clutter.

Dataset	Task	Clutter versions	Images
Shapes-VQA	Count four outline shapes	squiggles, blobs, look-alikes (broken or wavy)	2,000
Shapes-VQA Hard	Count shapes of one colour	wrong colour, look-alikes, misleading text, squiggles (4 levels), other shapes, all combined	5,000
Objects-VQA	Count apples, cats, dogs, cars	random objects, look-alikes, text, all combined	2,500
Lines-VQA	Follow a line to its box	other lines crossing it 0 to 8 times, coloured lines	3,000

3 Method

1. **Measure the effect (RQ1).** Run frozen VLMs on every version and compare with the clean version of each scene: Qwen3-VL-4B [5] as the main model, InternVL3.5, Molmo2 and Gemma 4 as checks.
2. **Show the entanglement (RQ2).** Targets are pixel-identical in the clean and cluttered image, so any change in their representation is leaked clutter. A probe tests whether it can be read back; swapping in the clean representation tests whether it causes errors.
3. **Disentangle (RQ3).** A sparse autoencoder (SAE) splits the image representation into many features. We switch off those that fire on clutter (found via the object locations) and let the same frozen model answer [4].

4 Evaluation and success criteria

Success means accuracy on cluttered images is clearly higher after the fix than before (it need not reach the clean level), measured as the share of lost accuracy that comes back: *gap closed* = $(\text{acc}_{\text{after}} - \text{acc}_{\text{cluttered}}) / (\text{acc}_{\text{clean}} - \text{acc}_{\text{cluttered}})$. **Controls:** switching off as many random features must not help, and clean accuracy must not drop. **Statistics:** 95% bootstrap intervals over scenes, McNemar tests. **Predictions:** largest gains for misleading text and unrelated objects, smallest for look-alikes, which share features with real targets.

5 Preliminary findings

Our first study confirms the accuracy drop: in all four datasets and every model tested, clutter lowered accuracy, often sharply, though the answer never changed. Look-alikes hurt most: on Objects-VQA the main model gets almost every clean image right, but fewer than a third with look-alikes. Models also copy misleading text and guess rather than trace lines, so Lines-VQA cannot test entanglement; Objects-VQA is the main test set.

6 Expected contributions and risks

Contributions: reusable clutter datasets with pixel-checked labels; a direct test of entanglement on targets whose pixels never change; a fix that needs no retraining of the VLM.

Risks: the information may be lost rather than mixed (then the limit is perception, and we report that); the leak may be harmless (then we switch off the clutter regions instead); look-alikes may share features with real targets, so switching them off could also remove real objects (we measure this trade-off).

References

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5. Qwen Team. Qwen3-VL-4B-Instruct model card. Hugging Face.